

Team Fish — Sensor Hardware Budget

FunkyForelle · CHF 22,000

Team Fish is responsible for sensor selection, procurement, calibration, and deployment across 7–8 monitoring sites on Swiss rivers. Based at OST Rapperswil (Lake Zurich campus). University lab resources (calibration equipment, 3D printers, IoT kits) used at no cost.

Item	Qty	Unit CHF	Total CHF
PILOT SITE — Patrick's river (full multi-parameter)			
In-Situ Aqua TROLL 500 (DO + temp + conductivity + depth)	1	3,500	3,500
Telemetry module (GSM) + weatherproof housing	1	800	800
Solar panel + battery backup + mounting hardware	1	800	800
SECONDARY SITES — DO + temperature			
YSI ProODO (DO + temp)	3	1,800	5,400
Telemetry module + weatherproof housing	3	700	2,100
Solar panel + battery	3	500	1,500
VALIDATION SITES — temperature + depth only			
Keller Series 36W (temp + depth logger)	3	500	1,500
Basic telemetry module	3	300	900
SHARED COSTS			
Field installation — travel, permits, 7 sites	1	2,000	2,000
SIM cards + data plans (7 sites × 4 months)	1	700	700
Consumables — cables, connectors, O-rings, desiccant	1	400	400
RESERVE			
Sensor failure / replacement contingency	1	1,400	1,400
TOTAL			21,000
University resources used at no cost (OST lab, 3D printer, calibration equipment)			(0)
Budget envelope			22,000
Remaining reserve			1,000

Notes

- OST Rapperswil IoT lab provides reference calibration equipment, prototyping components, and 3D printing at no cost to enrolled students.
- Patrick Hager (Fischereiverein See + Gaster) advises on site selection at no charge.
- All sensor hardware will be documented and open-sourced as part of the project deliverables.
- Prices are indicative (September 2026); final selection subject to market analysis by the intern.